

Sangsig Kim

professor

基本信息

姓名: Sangsig Kim

所属院校: 高丽大学

地区: South Korea

学科领域

微电子科学与工程

材料物理

纳米材料与技术

集成电路设计与集成系统

柔性电子学

材料科学与工程

研究领域

微电子器件与制造

薄膜材料与表面物理

纳米电子与先进工艺

半导体与光电子材料

柔性器件制备与工艺

纳米电子学与光子学

存储器设计与架构

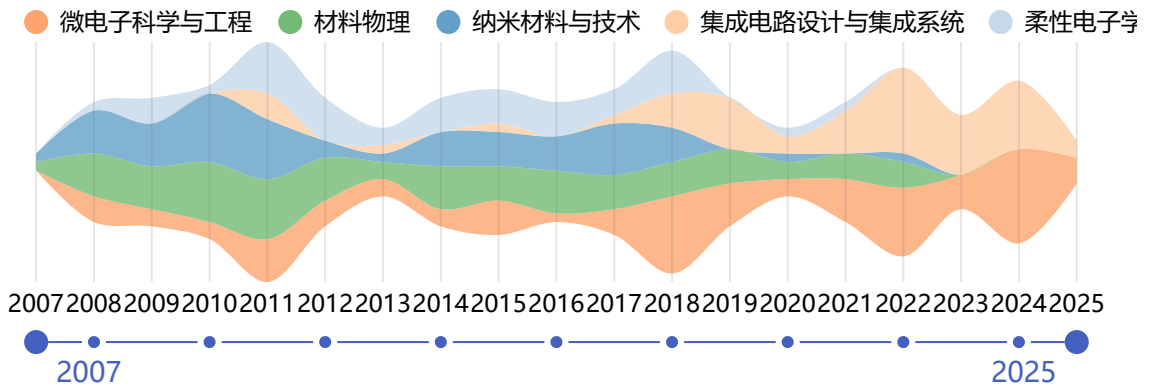
低功耗电路与能量管理

纳米合成与组装技术

个人简介

Sangsig Kim is a professor at the School of Electrical Engineering. With an h-index of 40 and 7690 citations, he has been engaged in research from 1992 to 2024. His active research topics include Nanowire, Field - Effect Transistor, Nanoparticle, etc. He has a significant research output, including 360 articles, 15 conference contributions, 7 conference articles, 3 papers, and 7 other types of works. In 2024, he co-authored multiple peer-reviewed journal articles in fields such as advanced electronic materials and scientific reports.

近年发表的论文



Reconfigurable Logic Gate Enabled by Dual-Gating of Silicon Nanomembrane Field-Effect Transistors

ADVANCED MATERIALS TECHNOLOGIES

Hwang, Joonha; Kwon, Jong Ik; Choi, Moon Kee; Lim, Jung Ah; Kim, Sangsig; Choi, Changsoon

被引用次数: 0

2025

Logic-in-memory cell enabling binary and ternary Boolean logics

SCIENCE CHINA-INFORMATION SCIENCES

Oh, Jeongyun; Jeon, Juhee; Shin, Yunwoo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2025

Stateful Full Adder Using Silicon Diodes

ADVANCED INTELLIGENT SYSTEMS

Son, Jaemin; Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2025

Binary and ternary compatible NAND/NOR logic-in-memory cell constructed with single-gated feedback field-effect transistors

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Lee, Donghyung; Shin, Yunwoo; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2025

Binarized Neural Network Comprising Quasi-Nonvolatile Memory Devices for Neuromorphic Computing

ADVANCED ELECTRONIC MATERIALS

Shin, Yunwoo; Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Bidirectional Synaptic Operations of Triple-Gated Silicon Nanosheet Transistors with Reconfigurable Memory Characteristics

ADVANCED ELECTRONIC MATERIALS

Shin, Yunwoo; Son, Jaemin; Jeon, Juhee; Ryu, Seungho; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Stateful Logic Operation of Gated Silicon Diodes for In-Memory Computing

ADVANCED ELECTRONIC MATERIALS

Son, Jaemin; Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 1

2024

Capacitorless Two-Transistor Dynamic Random-Access Memory Cells Comprising Amorphous Indium-Tin-Gallium-Zinc Oxide Thin-Film Transistors for the Multiply-Accumulate Operation

ADVANCED MATERIALS TECHNOLOGIES

Ryu, Seungho; Kang, Mingu; Cho, Kyoungah; Kim, Sangsig

被引用次数: 1

2024

Generation and Storage of Random Voltage Values via Ring Oscillators Comprising Feedback Field-Effect Transistors

NANOMATERIALS

Son, Jaemin; Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Temperature-Dependent Feedback Operations of Triple-Gate Field-Effect Transistors

NANOMATERIALS

Park, Taeho; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Read Operation Mechanism of Feedback Field-Effect Transistors with Quasi-Nonvolatile Memory States

NANOMATERIALS

Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Delay characteristics of quasi-nonvolatile memory devices operating in positive feedback mechanism

NANOTECHNOLOGY

Oh, Jeongyun; Jeon, Juhee; Shin, Yunwoo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Gate-bias stability of triple-gated feedback field-effect transistors with silicon nanosheet channels

NANOTECHNOLOGY

Heo, Hyojoo; Shin, Yunwoo; Son, Jaemin; Ryu, Seungho; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Effect of interface defects on electrical characteristics of a-ITGZO TFTs under bottom, top, and dual gatings

HELIYON

Kang, Mingu; Cho, Kyoungah; Seol, Minhyeok; Kim, Sangsub; Kim, Sangsig

被引用次数: 0

2024

Binary and ternary logic-in-memory using nanosheet feedback field-effect transistors with triple-gated structure

SCIENTIFIC REPORTS

Han, Jongseong; Son, Jaemin; Ryu, Seungho; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Binarized neural network of diode array with high concordance to vector-matrix multiplication

SCIENTIFIC REPORTS

Shin, Yunwoo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 0

2024

Reconfigurable Logic-In-Memory Cell Comprising Triple-Gated Feedback Field-Effect Transistors

ADVANCED ELECTRONIC MATERIALS

Han, Jongseong; Son, Jaemin; Jeon, Juhee; Shin, Yunwoo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 1

2023

Logic-In-Memory Characteristics of Reconfigurable Feedback Field-Effect Transistors with Double-Gated Structure

ADVANCED ELECTRONIC MATERIALS

Shin, Yunwoo; Son, Jaemin; Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 1

2023

Logic-in-Memory Operation of Ternary NAND/NOR Universal Logic Gates using Double-Gated Feedback Field-Effect Transistors

ADVANCED ELECTRONIC MATERIALS

Son, Jaemin; Shin, Yunwoo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 8

2023

Effects of Interface States on Electrical Characteristics of Feedback Field-Effect Transistors

IEEE ACCESS

Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 3

2023

Disturbance Characteristics of 1T DRAM Arrays Consisting of Feedback Field-Effect Transistors

MICROMACHINES

Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 4

2023

Prediction of Device Characteristics of Feedback Field-Effect Transistors Using TCAD-Augmented Machine Learning

MICROMACHINES

Woo, Sola; Jeon, Juhee; Kim, Sangsig

被引用次数: 3

2023

Effect of dual gating on electrical characteristics of amorphous indium-tin-gallium-zinc-oxide TFTs

ELECTRONICS LETTERS

Kong, Heesung; Cho, Kyoungah; Kang, Mingu; Kim, Jaybum; Lee, Sunhee; Lim, Junhyung; Kim, Sangsig

被引用次数: 2

2023

Performance Prediction of Hybrid Energy Harvesting Devices Using Machine Learning

ACS APPLIED MATERIALS & INTERFACES

Park, Yoonbeom; Cho, Kyoungah; Kim, Sangsig

被引用次数: 17

2022

Reconfigurable Logic-in-Memory Using Silicon Transistors

ADVANCED MATERIALS TECHNOLOGIES

Lim, Doohyeok; Cho, Kyoungah; Kim, Sangsig

被引用次数: 7

2022

Reconfiguration of operation modes in silicon nanowire field-effect transistors by electrostatic virtual doping

NANOTECHNOLOGY

Kim, Taekham; Lim, Doohyeok; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 4

2022

Temperature-Dependent Electrical Characteristics of p-Channel Mode Feedback Field-Effect Transistors

IEEE ACCESS

Park, Taeho; Lee, Jaehwan; Son, Jaemin; Jeon, Juhee; Shin, Yeonwoo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 1

2022

Annealing effect on thermoelectric characteristics of Spin-coated Cu₂Se nanoparticle thin films

CHEMICAL PHYSICS LETTERS

Lee, Jaehwan; Cho, Kyoungah; Park, Taeho; Kim, Sangsig

被引用次数: 3

2022

Bias-stress stability of top-gate coplanar a-ITGZO TFTs with HfO₂ and HfAlO gate dielectrics

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Kong, Heesung; Cho, Kyoungah; Lee, Hosang; Lee, Seungjun; Lim, Junhyung; Kim, Sangsig

被引用次数: 10

2022

Design and Simulation of Logic-In-Memory Inverter Based on a Silicon Nanowire Feedback Field-Effect Transistor

MICROMACHINES

Baek, Eunwoo; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 7

2022

Dependence of latch-up and threshold voltages on channel length in single-gated feedback field-effect transistor

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Woo, Sola; Kim, Sangsig

被引用次数: 2

2022

Logic and memory characteristics of an inverter comprising a feedback FET and a MOSFET

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lim, Eunhyeok; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 3

2022

Performance Enhancement of Hybrid Energy Devices Using Cooling Patches

INTERNATIONAL JOURNAL OF PHOTOENERGY

Lee, Jaehwan; Cho, Kyoungah; Park, Yoonbeom; Park, Sungeun; Song, Hee-eun; Kim, Sangsig

被引用次数: 2

2022

Universal logic-in-memory cell enabling all basic Boolean algebra logic

SCIENTIFIC REPORTS

Baek, Eunwoo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 4

2022

New ternary inverter with memory function using silicon feedback field-effect transistors

SCIENTIFIC REPORTS

Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 6

2022

Logic and memory functions of an inverter comprising reconfigurable double gated feedback field effect transistors

SCIENTIFIC REPORTS

Jeon, Juhee; Woo, Sola; Cho, Kyoungah; Kim, Sangsig

被引用次数: 4

2022

NAND and NOR logic-in-memory comprising silicon nanowire feedback field-effect transistors

SCIENTIFIC REPORTS

Yang, Yejin; Jeon, Juhee; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 5

2022

Neural oscillation of single silicon nanowire neuron device with no external bias voltage

SCIENTIFIC REPORTS

Woo, Sola; Kim, Sangsig

被引用次数: 7

2022

Ammonia Gas Sensing Properties of 6,13-Bis(tri-isopropylsilyl)ethynyl Pentacene Based Field-Effect Transistor

TRANSACTIONS ON ELECTRICAL AND ELECTRONIC MATERIALS

Lee, Byeong Hyeon; Kim, Sangsig; Lee, Sang Yeol

被引用次数: 10

2022

Electrical Performance of Amorphous Oxide/Colloidal Quantum Dot/Amorphous Oxide Hybrid Thin Film Transistor

TRANSACTIONS ON ELECTRICAL AND ELECTRONIC MATERIALS

Lee, Byeong Hyeon; Cho, Kyung-Sang; Kim, Sangsig; Lee, Sang Yeol

被引用次数: 0

2022

Electrical characteristics of flexible amorphous indium-tin-gallium-zinc oxide thin-film transistors under repetitive mechanical stress

JAPANESE JOURNAL OF APPLIED PHYSICS

Lee, Hosang; Cho, Kyoungah; Kong, Heesung; Lee, Seungjun; Lim, Junhyung; Kim, Sangsig

被引用次数: 5

2021

Thin-Film Thermoelectric Generators Comprising Molybdenum-Based MXenes pn Modules

ADVANCED MATERIALS TECHNOLOGIES

Park, Taeho; Cho, Kyoungah; Kim, Sangsig

被引用次数: 16

2021

Inverting logic-in-memory cells comprising silicon nanowire feedback field-effect transistors

NANOTECHNOLOGY

Park, Young-Soo; Lim, Doohyeok; Son, Jaemin; Jeon, Juhee; Cho, Kyoungah; Kim, Sangsig

被引用次数: 16

2021

Electrical Stability of p-Channel Feedback Field-Effect Transistors Under Bias Stresses

IEEE ACCESS

Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 2

2021

Single Silicon Neuron Device Enabling Neuronal Oscillation and Stochastic Dynamics

IEEE ELECTRON DEVICE LETTERS

Lim, Doohyeok; Cho, Kyoungah; Kim, Sangsig

被引用次数: 19

2021

Vertical all-in-one energy systems constructed with thermoelectric generators and microsupercapacitors

JOURNAL OF POWER SOURCES

Park, Yoonbeom; Cho, Kyoungah; Kim, Sangsig

被引用次数: 7

2021

ASPICEmodel of p-channel silicon tunneling field-effect transistors for logic applications

INTERNATIONAL JOURNAL OF NUMERICAL MODELLING-ELECTRONIC NETWORKS DEVICES AND FIELDS

Woo, Sola; Jeon, Juhee; Kim, Sangsig

被引用次数: 1

2021

Integrate-and-Fire Neuron Circuit Without External Bias Voltages

FRONTIERS IN NEUROSCIENCE

Park, Young-Soo; Woo, Sola; Lim, Doohyeok; Cho, Kyoungah; Kim, Sangsig

被引用次数: 13

2021

Simulation studies on electrical characteristics of silicon nanowire feedback field-effect transistors with interface trap charges

SCIENTIFIC REPORTS

Yang, Yejin; Park, Young-Soo; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 3

2021

One-transistor static random-access memory cell array comprising single-gated feedback field-effect transistors

SCIENTIFIC REPORTS

Choi, Sangik; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 5

2021

Effect of Silicon Doping on the Electrical Performance of Amorphous SiInZnO Thin-film Transistors (vol 22, pg 133, 2021)

TRANSACTIONS ON ELECTRICAL AND ELECTRONIC MATERIALS

Lee, Byeong Hyeon; Kim, Dae-Hwan; Lee, Doo-Yong; Park, Sungkyun; Kim, Sangsig; Kwon, Hyuck-In; Lee, Sang Yeol

被引用次数: 0

2021

Effect of Silicon Doping on the Electrical Performance of Amorphous SiInZnO Thin-film Transistors

TRANSACTIONS ON ELECTRICAL AND ELECTRONIC MATERIALS

Lee, Byeong Hyeon; Kim, Dae-Hwan; Lee, Doo-Yong; Park, Sungkyun; Kim, Sangsig; Kwon, Hyuck-In; Lee, Sang Yeol

被引用次数: 3

2021

Device design of single-gated feedback field-effect transistors to achieve latch-up behaviors with high current gains

CURRENT APPLIED PHYSICS

Woo, Sola; Kim, Sangsig

被引用次数: 8

2020

Energy devices generating and storing electricity from finger and solar thermal energy

NANO ENERGY

Yang, Seunggen; Cho, Kyoungah; Kim, Sangsig

被引用次数: 13

2020

Performance of Hybrid Energy Devices Consisting of Photovoltaic Cells and Thermoelectric Generators

ACS APPLIED MATERIALS & INTERFACES

Park, Yoonbeom; Cho, Kyoungah; Yang, Seunggen; Park, Taeho; Park, Sungeun; Song, Hee-eun; Kim, Soo Mm; Kim, Sangsig

被引用次数: 9

2020

Quasi-Nonvolatile Silicon Memory Device

ADVANCED MATERIALS TECHNOLOGIES

Lim, Doohyeok; Son, Jaemin; Cho, Kyoungah; Kim, Sangsig

被引用次数: 30

2020

Amorphous oxide based microcavity color filter with high selectivity

OPTICAL MATERIALS

Lee, Byeong Hyeon; Hwang, Jin Young; Kim, Sangsig; Lee, Sang Yeol

被引用次数: 1

2020

Implementation and Characterization of an Integrate-and-Fire Neuron Circuit Using a Silicon Nanowire Feedback Field-Effect Transistor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Woo, Sola; Cho, Jinsun; Lim, Doohyeok; Park, Young-Soo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 21

2020

Enhanced Thermoelectric Characteristics of Ag₂Se Nanoparticle Thin Films by Embedding Silicon Nanowires

ENERGIES

Yang, Seunggen; Cho, Kyoungah; Kim, Sangsig

被引用次数: 1

2020

Electrical characteristics of bendable a-ITGZO TFTs on colorless polyimide substrates

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lee, Hosang; Cho, Kyoungah; Kim, Donghyun; Kim, Sangsig

被引用次数: 6

2020

Investigation on trap density depending on Si ratio in amorphous SiZnSnO thin-film transistors

PHYSICA B-CONDENSED MATTER

Lee, Byeong Hyeon; Hong, Sae-Young; Kim, Dae-Hwan; Kim, Sangsig; Kwon, Hyuck-In; Lee, Sang Yeol

被引用次数: 4

2019

Investigation on the improvement of stability of nitrogen doped amorphous SiInZnO thin-film transistors

SOLID-STATE ELECTRONICS

Lee, Byeong Hyeon; Lee, Doo-Yong; Lee, Ji Ye; Park, Sungkyun; Kim, Sangsig; Lee, Sang Yeol

被引用次数: 6

2019

Investigation on dependency mechanism of inverter voltage gain on current level of photo stressed depletion mode thin-film transistors

SOLID-STATE ELECTRONICS

Lee, Byeong Hyeon; Kim, Sangsig; Lee, Sang Yeol

被引用次数: 4

2019

A laterally designed all-in-one energy device using a thermoelectric generator-coupled micro supercapacitor

NANO ENERGY

Yang, Kyungwhan; Cho, Kyoungah; Yang, Seunggen; Park, Yoonbeom; Kim, Sangsig

被引用次数: 32

2019

Polarity control of carrier injection for nanowire feedback field-effect transistors

NANO RESEARCH

Lim, Doohyeok; Kim, Sangsig

被引用次数: 21

2019

Transposable 3T-SRAM Synaptic Array Using Independent Double-Gate Feedback Field-Effect Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Woo, Sola; Cho, Jinsun; Lim, Doohyeok; Cho, Kyoungah; Kim, Sangsig

被引用次数: 16

2019

Nonvolatile and Volatile Memory Characteristics of a Silicon Nanowire Feedback Field-Effect Transistor With a Nitride Charge-Storage Layer

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kang, Hyungu; Cho, Jinsun; Kim, Yoonjoong; Lim, Doohyeok; Woo, Sola; Cho, Kyoungah; Kim, Sangsig

被引用次数: 18

2019

Static Random Access Memory Characteristics of Single-Gated Feedback Field-Effect Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Cho, Jinsun; Lim, Doohyeok; Woo, Sola; Cho, Kyoungah; Kim, Sangsig

被引用次数: 30

2019

Covered Source-Channel Tunnel Field-Effect Transistors With Trench Gate Structures

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Woo, Sola; Kim, Sangsig

被引用次数: 12

2019

Optically tunable feedback operation of silicon nanowire transistors

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Lim, Doohyeok; Kim, Sangsig

被引用次数: 2

2019

Investigation on energy bandgap states of amorphous SiZnSnO thin films

SCIENTIFIC REPORTS

Lee, Byeong Hyeon; Cho, Kyung-Sang; Lee, Doo-Yong; Sohn, Ahum; Lee, Ji Ye; Choo, Hyuck; Park, Sungkyun; Kim, Sang-Woo; Kim, Sangsig; Lee, Sang Yeol

被引用次数: 24

2019

Mechanism of carrier controllability with metal capping layer on amorphous oxide SiZnSnO semiconductor

SCIENTIFIC REPORTS

Lee, Byeong Hyeon; Sohn, Ahrum; Kim, Sangsig; Lee, Sang Yeol

被引用次数: 32

2019

Electrical characteristics of silicon nanowire CMOS inverters under illumination

OPTICS EXPRESS

Yoo, Jeuk; Kim, Yoonjoong; Lim, Doohyeok; Kim, Sangsig

被引用次数: 6

2018

Performance of ring oscillators composed of gate-all-around FETs with varying numbers of nanowire channels using TCAD simulation

CURRENT APPLIED PHYSICS

Kim, Suta; Kim, Minsuk; Woo, Sola; Kang, Hyungu; Kim, Sangsig

被引用次数: 7

2018

Bendable thermoelectric generators composed of p- and n-type silver chalcogenide nanoparticle thin films

NANO ENERGY

Yang, Seunggen; Cho, Kyoungah; Park, Yoonbeom; Kim, Sangsig

被引用次数: 27

2018

Silicon nanowire ratioed inverters on bendable substrates

NANO RESEARCH

Moon, Jeongje; Kim, Yoonjoong; Lim, Doohyeok; Im, Kyeungmin; Kim, Sangsig

被引用次数: 2

2018

Silicon nanowire CMOS NOR logic gates featuring one-volt operation on bendable substrates

NANO RESEARCH

Moon, Jeongje; Kim, Yoonjoong; Lim, Doohyeok; Kim, Sangsig

被引用次数: 7

2018

Switchable-Memory Operation of Silicon Nanowire Transistor

ADVANCED ELECTRONIC MATERIALS

Kim, Yoonjoong; Cho, Jinsun; Lim, Doohyeok; Woo, Solo; Cho, Kyoungah; Kim, Sangsig

被引用次数: 12

2018

Identifying DNA mismatches at single-nucleotide resolution by probing individual surface potentials of DNA-capped nanoparticles

NANOSCALE

Lee, Hyungbeen; Lee, Sang Won; Lee, Gyudo; Lee, Wonseok; Nam, Kihwan; Lee, Jeong Hoon; Hwang, Kyo Seon; Yang, Jaemoon; Lee, Hyeyoung; Kim, Sangsig; Lee, Sang Woo; Yoon, Dae Sung

Photocurrent Engineering of Silicon Nanowire Field-Effect Transistors by Ultrathin Poly(3-hexylthiophene)

ADVANCED MATERIALS INTERFACES

In, Chihun; Kim, Daewon; Roh, Young-Geun; Kim, Sang Won; Lee, Hyangsook; Park, Yeonsang; Kim, Sangsig; Kim, Un Jeong; Choi, Hyunyoung; Hwang, Sung Woo

被引用次数: 2

2018

Feedback and tunneling operations of a p+-i-n+ silicon nanowire field-effect transistor

NANOTECHNOLOGY

Kim, Yoonjoong; Lim, Doohyeok; Cho, Jinsun; Kim, Sangsig

被引用次数: 6

2018

Sol-Gel Processed p-Type CuO Phototransistor for a Near-Infrared Sensor

IEEE ELECTRON DEVICE LETTERS

Lee, Sojeong; Lee, Won-Yong; Jang, Bongho; Kim, Taegyun; Bae, Jin-Hyuk; Cho, Kyoungah; Kim, Sangsig; Jang, Jaewon

被引用次数: 50

2018

Nondestructive Readout Memory Characteristics of Silicon Nanowire Biristors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lim, Doohyeok; Kim, Minsuk; Kim, Yoonjoong; Cho, Jinsun; Kim, Sangsig

被引用次数: 9

2018

Lithography-free fabrication of field effect transistor channels with randomly contact-printed black phosphorus flakes

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Yoo, Seolhee; Kim, Sangsig; Song, Yong-Won

被引用次数: 3

2018

A SPICE model of silicon tunneling field-effect transistors

MICROELECTRONIC ENGINEERING

Woo, Sola; Kim, Minsuk; Kim, Sangsig

被引用次数: 3

2018

Effect of substrates on the electrical characteristics of a silicon nanowire feedback field-effect transistor under bending stresses

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kim, Yoonjoong; Kim, Sangsig

被引用次数: 3

2018

Effect of stiffness modulation on mechanical stability of stretchable a-IGZO TFTs

SUPERLATTICES AND MICROSTRUCTURES

Park, Hyungjin; Cho, Kyoungah; Oh, Hyungon; Kim, Sangsig

被引用次数: 3

2018

Bending-strain-induced localized density of states in amorphous indium-gallium-zinc-oxide thin-film transistors

SUPERLATTICES AND MICROSTRUCTURES

Woo, Sola; Kim, Minsuk; Oh, Hyungon; Cho, Kyoungah; Kim, Sangsig

被引用次数: 1

2018

Suppression of threshold voltage variability in MOSFETs by adjustment of ion implantation parameters

SUPERLATTICES AND MICROSTRUCTURES

Park, Jae Hyun; Chang, Tae-sig; Kim, Minsuk; Woo, Sola; Kim, Sangsig

被引用次数: 1

2018

Electrochemical characteristics of flexible micro supercapacitors with reduced graphene oxide-carbon nanotubes composite electrodes

SUPERLATTICES AND MICROSTRUCTURES

Yang, Kyungwhan; Cho, Kyoungah; Kim, Sangsig

被引用次数: 10

2018

Effect of physical densification on sub-gap density of states in amorphous InGaZnO thin films

SUPERLATTICES AND MICROSTRUCTURES

Oh, Hyungon; Cho, Kyoungah; Kim, Sangsig

被引用次数: 3

2018

Electrical characteristics of a bendable a-Si:H thin film transistor with overlapped gate and source/drain regions

APPLIED PHYSICS LETTERS

Oh, Hyungon; Cho, Kyoungah; Kim, Sangsig

被引用次数: 5

2017

Large Voltage Generation of Flexible Thermoelectric Nanocrystal Thin Films by Finger Contact

ADVANCED ENERGY MATERIALS

Choi, Jinyong; Cho, Kyoungah; Yun, Junggwon; Park, Yoonbeom; Yang, Seunggen; Kim, Sangsig

被引用次数: 34

2017

Flexible Thermoelectric Generators Composed of n-and p-Type Silicon Nanowires Fabricated by Top-Down Method

ADVANCED ENERGY MATERIALS

Choi, Jinyong; Cho, Kyoungah; Kim, Sangsig

被引用次数: 33

2017

Thermoelectric characteristics of nanocomposites made of HgSe and Ag nanoparticles for flexible thermoelectric devices

NANO RESEARCH

Yun, Junggwon; Cho, Kyoungah; Park, Yoonbeom; Yang, Seunggen; Choi, Jinyong; Kim, Sangsig

被引用次数: 7

2017

Thermoelectric characteristics of glass fibers coated with ZnO and Al-doped ZnO

MATERIALS RESEARCH BULLETIN

Park, Yoonbeom; Cho, Kyoungah; Kim, Sangsig

被引用次数: 10

2017

Mechanical strain-induced defect states in amorphous silicon channel layers of thin-film transistors

THIN SOLID FILMS

Kim, Minsuk; Oh, Hyungon; Park, Sukhyung; Cho, Kyoungah; Kim, Sangsig

被引用次数: 3

2017

Thermoelectric characteristics of γ -Ag₂Te nanoparticle thin films on flexible substrates

THIN SOLID FILMS

Yang, Seunggen; Cho, Kyoungah; Yun, Junggwon; Choi, Jinyong; Kim, Sangsig

被引用次数: 19

2017

Steep switching characteristics of single-gated feedback field-effect transistors

NANOTECHNOLOGY

Kim, Minsuk; Kim, Yoonjoong; Lim, Doohyeok; Woo, Sola; Cho, Kyoungah; Kim, Sangsig

被引用次数: 45

2017

Channel width dependence of electrical characteristics of a-Si:H TFTs under bending stresses

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Oh, Hyungon; Cho, Kyoungah; Kim, Sangsig

被引用次数: 4

2017

Effect of Si on the Energy Band Gap Modulation and Performance of Silicon Indium Zinc Oxide Thin-Film Transistors

SCIENTIFIC REPORTS

Choi, Jun Young; Heo, Keun; Cho, Kyung-Sang; Hwang, Sung Woo; Chung, JaeGwan; Kim, Sangsig; Lee, Byeong Hyeon; Lee, Sang Yeol

被引用次数: 32

2017

Memory characteristics of silicon nanowire transistors generated by weak impact ionization

SCIENTIFIC REPORTS

Lim, Doohyeok; Kim, Minsuk; Kim, Yoonjoong; Kim, Sangsig

被引用次数: 17

2017

Bendable solid-state supercapacitors with Au nanoparticle-embedded graphene hydrogel films

SCIENTIFIC REPORTS

Yang, Kyungwhan; Cho, Kyoungah; Yoon, Dae Sung; Kim, Sangsig

Electrical and mechanical characteristics of fully transparent IZO thin-film transistors on stress-relieving bendable substrates

APPLIED PHYSICS LETTERS

Park, Sukhyung; Cho, Kyoungah; Oh, Hyungon; Kim, Sangsig

被引用次数: 17

2016

Nanowatt power operation of silicon nanowire NAND logic gates on bendable substrates

NANO RESEARCH

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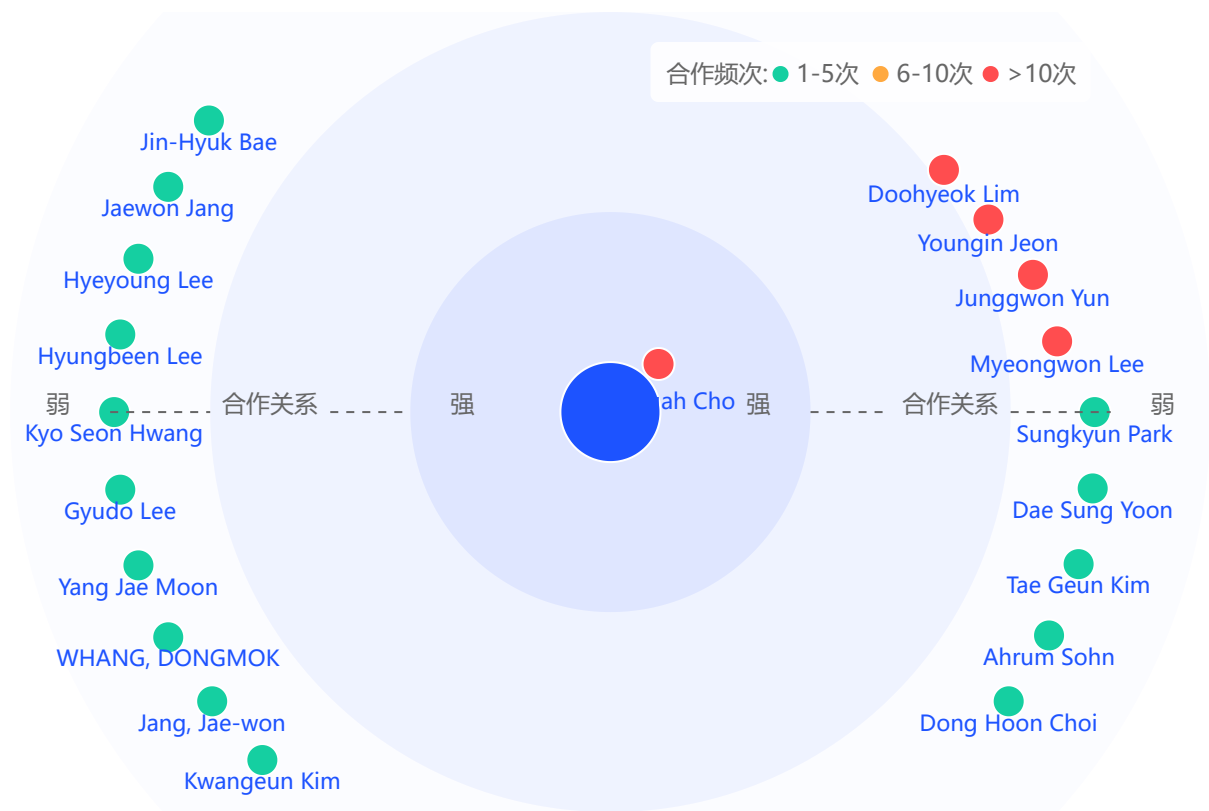
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